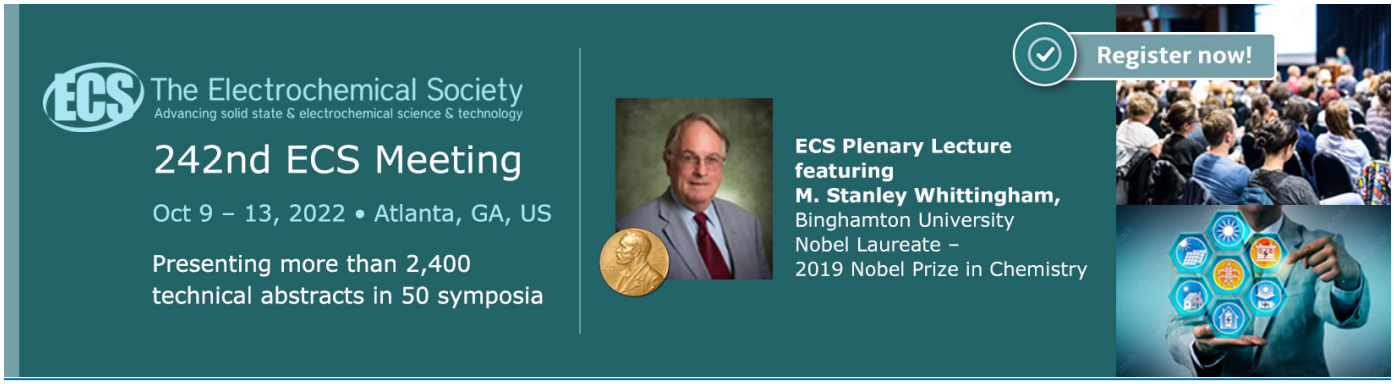
ECS Transactions

The Electrochemical Society
Advancing solid state & electrochemical science & technology

A Novel High Temperature Metal - Air Battery

To cite this article: Wolfgang W Drenckhahn *et al* 2013 *ECS Trans.* **50** 125

View the article online for updates and enhancements.



This content was downloaded from IP address 130.215.123.157 on 15/09/2022 at 22:01

ECS Transactions, 50 (45) 125-135 (2013)
10.1149/05045.0125ecst ©The Electrochemical Society

A Novel High Temperature Metal–Air Battery

W. Drenckhahn[a], H. Greiner[c], M. Kühne[a], H. Landes[c], A. Leonide[c], K. Litzinger[b], C. Lu[b], C. Schuh[d] , J. Shull[b], and T. Soller[a]

[a] Siemens Energy Sector, 91058 Erlangen, Germany
[b] Siemens Energy Inc., 1310 Beulah Road – Churchill, Pittsburgh, PA 15235 – 5098, USA
[c] Siemens Corporate Technology, 91058 Erlangen, Germany
[d] Siemens Corporate Technology, 80200 Munich, Germany

Based on the planar SOFC and SOE technology, a high temperature metal–air battery was operated at temperatures between 700°C and 800°C. It reached charging and discharging capacities of about 2 hours at power densities of more than 250mW/cm². A Siemens stack concept was developed and validated in numerous lab experiments as well as in several stack demonstrators of more than 1kW power output.

The storage material is a proprietary development consisting of a iron/iron oxide combination which can be produced economically. It showed very low degradation rates of 1 - 2% in 1,000 charging / discharging cycles. More than 10,000 cycles were reached in 10x10 cm² short stacks. System efficiencies for the upper MW range of > 60% will be achieved.

1. Introduction

Siemens has many years of experience in the development of fuel cells, and is nowadays active in producing PEM-fuel cell systems for maritime application and produced several SOFC-plants in the 100 kW range. A decision was made in 2009 to channel the expertise gathered over many years in electrochemistry, plant and systems engineering into the development of an innovative energy storage device capable of serving the market confronted with the volatile fluctuations of regenerative power generation as well as providing support to and supplementing gas turbine based power plants.

Goal of the project was defined for this market to develop a novel high temperature Metal-Air Battery in the 100 MW range at target costs of 500-1000 € / kW and a discharge time of 2 hours. The cycle efficiency of the system should be better than 60% and 10,000 load cycles (this means, 10 years lifetime, 3 cycles / day) have to be demonstrated.

2. Basics

It was proposed to develop a high-temperature rechargeable oxide-ion battery (ROB) based on the well-known SOFC technology. The battery appeared promising in terms of its considerably simpler configuration compared to a fuel cell, because it does not require hydrogen or methane as a fuel gas. It just works with a stagnant atmosphere of a steam hydrogen mixture in contact with a proprietary storage material on the negative electrode side. The rechargeable oxide - ion battery is thus technically simpler than a fuel cell and has the advantage over other commercially available batteries that it is environment-friendly, inherent safe, low in cost and shows a competitive energy density.

The original idea was to fabricate a thick negative electrode in the order of 5 – 10mm in which the storage material is embedded. The storage medium was to be a metal that is electrochemically oxidized during discharge and again reduced to metal state when recharged.

In each case, coupling of electric current and oxygen transport across the cell is achieved by means of the solid electrolyte (for example, yttrium-stabilized ZrO_2) with the direction of flux depending on the difference between OCV and applied clamp voltage, see Fig. 1.

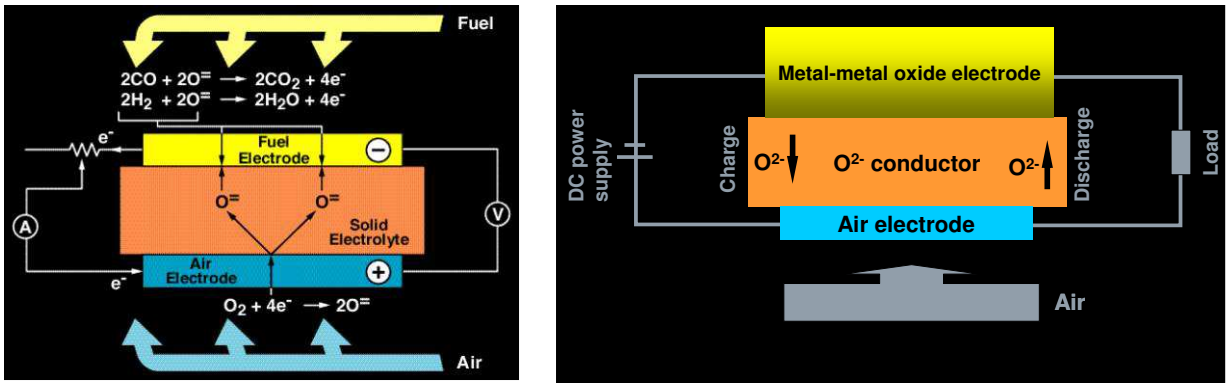


Figure 1. Functional Principle of SOFC (left side) and Functional Principle of ROB (right).

This first idea considering an electrically conducting storage electrode failed with respect to the required high power density due to the very low ionic and electronic bulk conductivities of the storage material and the high degradation of contacts upon cycling due to the reaction volume of the redox reaction. To find a solid material with sufficient transport properties some 100 compositions have been investigated.

Nevertheless, the relatively high resistances indicate that the current densities which could be achieved are insufficient for a technically expedient and economically justifiable scale. Transport by diffusion of molecular oxygen within a fluid filling the open pores around the metal particles had to be excluded as development path due to the extremely low oxygen concentration of $p(O_2)<10^{-20}$ bar at the reducing conditions in the negative electrode compartment.

However, this problem of low oxygen concentration can be resolved by the introduction of a gaseous redox pair with components of atmospheric pressure but which is thermodynamically equivalent to oxygen of the above low concentration, see (1) for more details. Fortunately a close to 1:1 mixture of hydrogen water-vapor has this feature if iron is taken as storage material at the operating temperature:

$$H_2 + ½\ O_2 = H_2O \qquad [1]$$

This mechanism thermodynamically achieves the same low oxygen activity on the surface of the Fe/FeO particles and the surface of the electrolyte, while concentrations of diffusing H_2 and H_2O molecules are kept high. A schematic illustration of the oxygen transport is shown in Fig. 2.

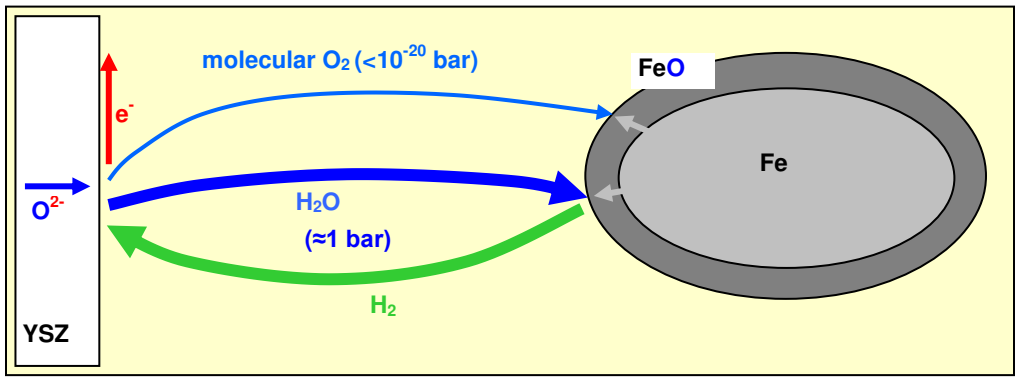


Figure 2. Principle of transport of oxygen by an H_2/H_2O redox shuttle (real gas pressure) in comparison to transport by oxygen molecules (theoretical pressure, calculated).

The mechanism is described here based on actual battery operation (Fig. 3):

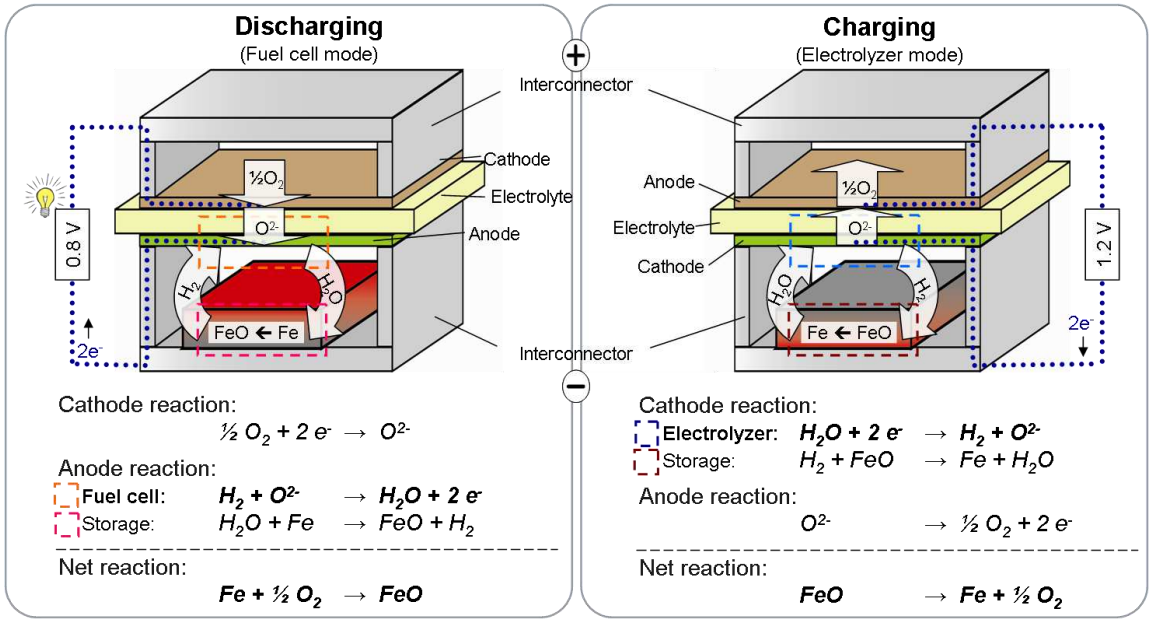


Figure 3. Combination of proven fuel cell / electrolyzer technology with metal/metal-oxide storage concept.

The following holds in zero-load operation (OCV). As long as there is a Fe/FeO mixture, the ratio of H_2O/H_2 is approximately 1, the oxygen chemical potential is equally high in the gas at the negative electrode and in the gas in the pores of the Fe/FeO mixture. On discharging, there is an external current flow from the positive electrode to the negative electrode, i.e. O^{2-} ions flow to the negative electrode and form H_2O with the hydrogen present there. The H_2O/H_2 ratio is now higher at the negative electrode than in the Fe/FeO pore system, with the result that H_2O diffuses into these pores and H_2 diffuses out. This increases the oxygen potential above that of Fe/FeO and oxidizes additional Fe. The continuous transfer of oxygen between negative electrode and Fe/FeO storage material employing H_2O and H_2 molecules represents a "shuttle mechanism".

During charging, there is an external current flow from the negative electrode to the positive electrode, i.e., O^{2-} ions are extracted from the negative electrode. The oxygen chemical potential is lowered at the negative electrode and the shuttle transfers this effect to the FeO, which is reduced to Fe.

3. Development of storage material

The development of a storage material capable of redox operation by the illustrated shuttle mechanism was the main effort for the new Siemens metal-air battery.
The following requirements have to be fulfilled:

- Reversible redox behavior of storage metal,
- broad operation temperature range,
- high volumetric charge density,
- low volumetric resistance with respect to the shuttle reaction,
- long cycle life,
- chemical compatibility with other cell components,
- cost effective raw materials and manufacturing processes.

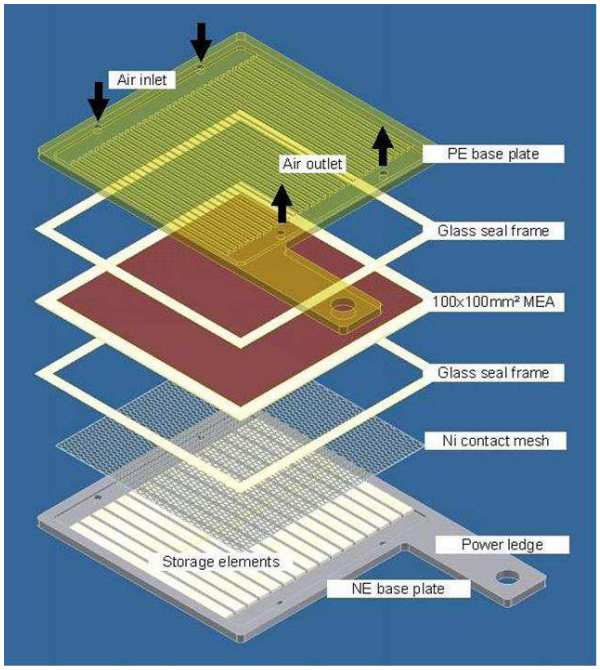
In order to identify the most suitable storage material, all in all more than 100 compositions were screened both via high throughput experimentation (utilizing a design of experiments approach) and conventional mixed oxide synthesis. All fabricated samples were characterized by annealing at operation temperature in operation atmospheres. Microstructure analysis and TGA measurements completed the pre-tests. Final tests of storage material always have to be done through electrochemical analysis performed by short stack tests as described in the next chapter.

In addition to the material development, also different production methods for bar-shaped storage elements were evaluated in order to ensure meeting the defined cost and performance goals. The investigated processes comprised tape casting, extrusion and uniaxial pressing. While the feasibility of producing high-quality storage elements could be successfully demonstrated with all three methods, uniaxial pressing was the most cost-competitive process.

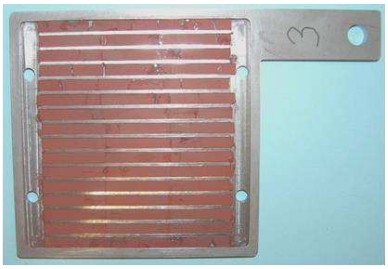
4. Short stack and stack development

A short stack is the smallest typical configuration of a stack (Fig. 4) and consists of the following components:

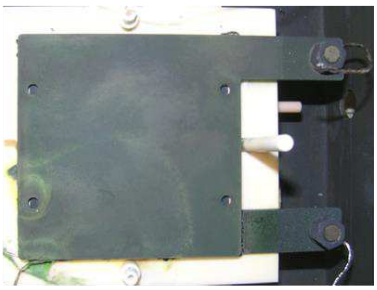
- a metallic base plate containing the deep channels required for implementation of the storage material. For construction with ESC cells, the proven material CYF (CrFe5Y) procured from Plansee is used. Crofer® 22 APU material from ThyssenKrupp is applied with ASC cells;
- a nickel grid to enhance the contact to the negative electrode,
- a membrane electrolyte assembly (MEA), consisting of the negative electrode, electrolyte and positive electrode and
- a metallic cover plate in which the channels for air (oxygen) supply are incorporated;
- frames of glass to gas tightly seal MEA, base plate and top plate.



Short stack, exploded view



Base plate with bars of storage material



Short stack (10x10cm²) ready for operation

Example of experimental set up for ESC, similar design for ASC

Figure 4. Short Stack arrangement (100/1, ESC).

The purpose of the short stack tests was as follows:

- Testing energy storage under conditions close to operating conditions,
- optimization of the storage material for performance and service life under comparable operating conditions,
- qualification of various manufacturing processes for the storage material.

The short stack tests were performed with ESC MEAs from Kerafol and ASC MEAs from FZ Jülich and SOFCpower. Target performance required could be demonstrated in several short stack tests. Fig. 5 shows that the performance goal of more than 250 mW/cm^2 power density was reached over 100 cycles and 1 hour discharge time. Note that the curves of cycle 1, 21, 41, 81 and 101 show nearly the same trace or even improved; means no degradation was observable during the test.

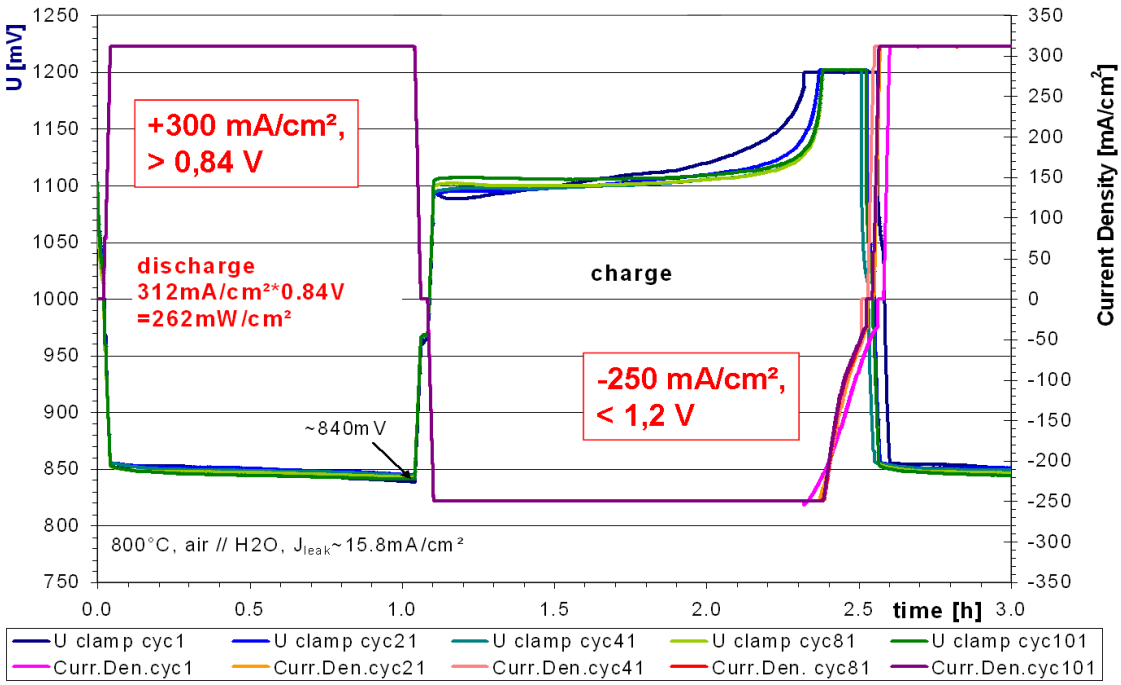


Figure 5. Short stack performance over 100 cycles exceeding 250 mW/cm^2 of discharging power.

In the same test a 2 hour discharge at 300 mA/cm^2 could also be demonstrated (Fig. 6). These tests were carried out by using ASC MEAs because of their higher power density. The power limiting factor is neither the shuttle or the storage material but still the MEA kinetics.

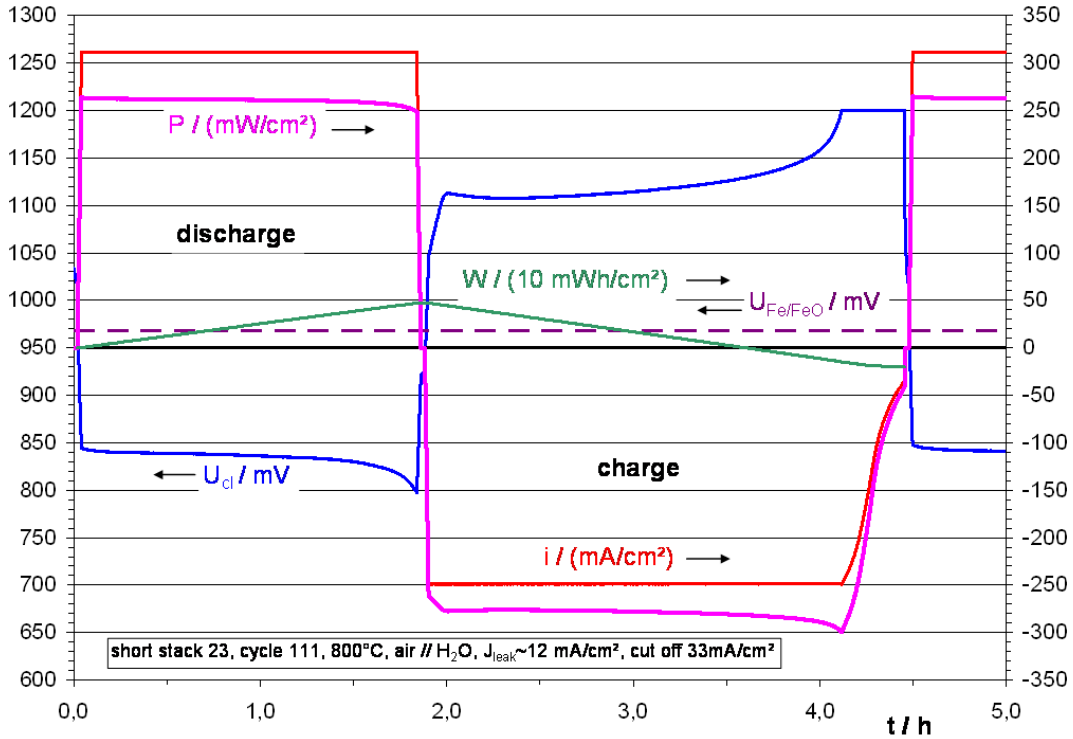


Figure 6. Demonstration of 2hrs discharge time@250mW/cm² (axes as in Fig. 5).

A further goal of the development was the construction, build up and test of stacks in the 1 kW class. At the beginning ESC cells from Kerafol and CFY interconnectors from Plansee were used to build up a 40 layer stack with 1 MEA / layer. 1 kW output was easily reached. The stack was operated over 70 charging cycles. By this test the feasibility of power scale-up was successfully demonstrated. Further on a 1kW stack based on ASC MEAs and Crofer® interconnectors plates was tested in the new Siemens multiple array design (Fig 7).

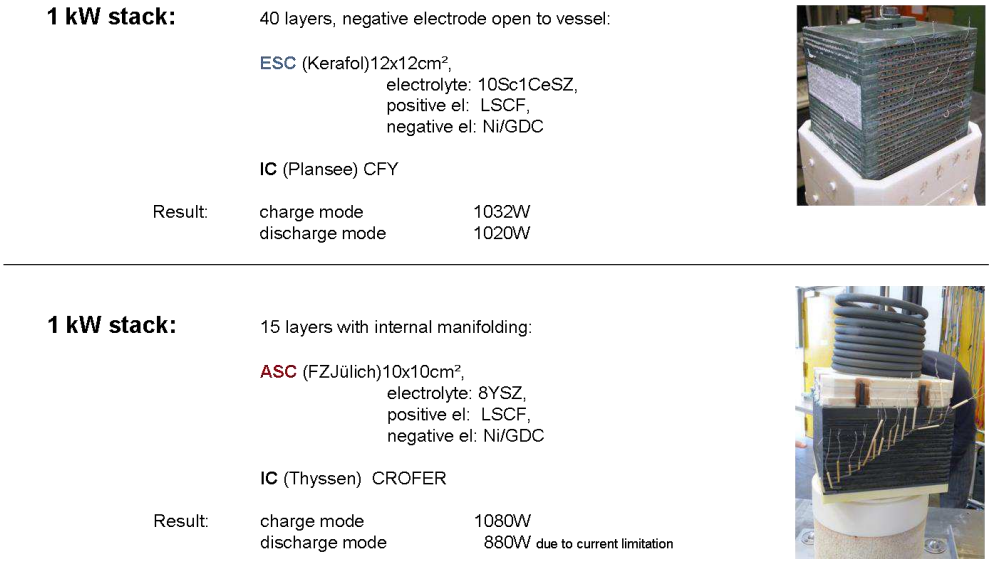


Figure 7. 1 kW stack tests based on ESC and ASC. Both set-ups reached the development target. Crofer® is a registered Trademark of ThyssenKrupp.

For the application of ASC MEAs, an own Siemens multiple cell array design stack was realized which allows the enlarging of the active area of each stack layer (Fig. 8).

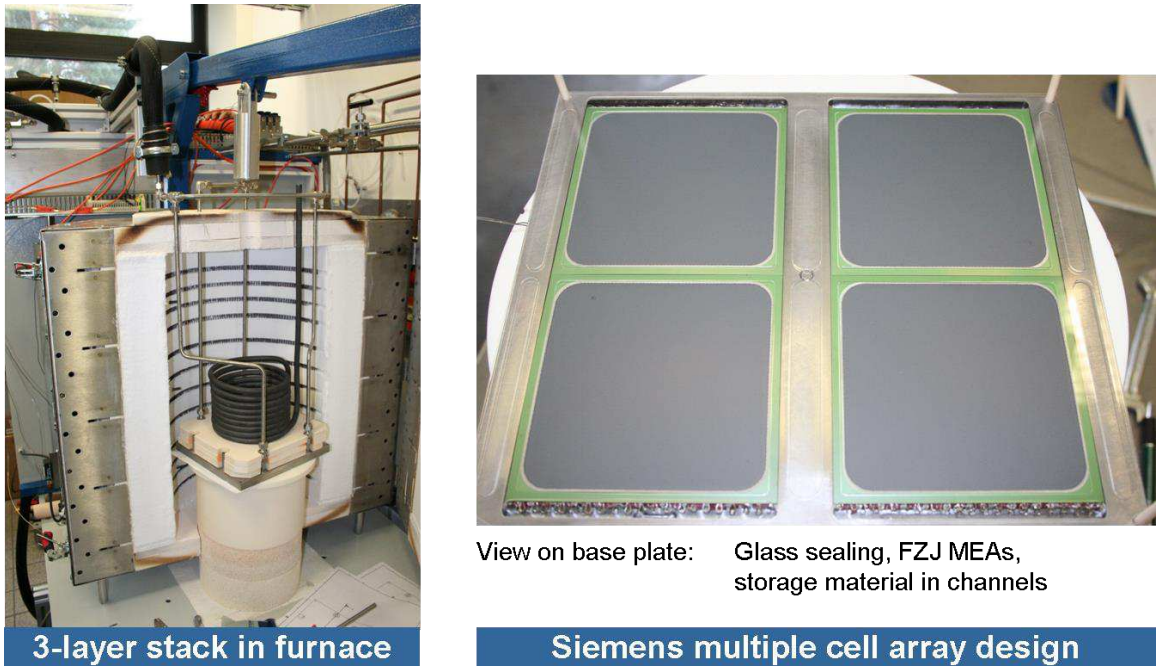


Figure 8. New Siemens stack design.

5. Long term stability

A battery has to operate reliably for several thousand charging/discharging cycles. For this development it was required to demonstrate 10,000 cycles. Two short stacks were built up consisting of ESC MEAs and CFY interconnectors. To eliminate the influence of aging of the positive electrode caused by volatile chromium the top plate was made out of gold (Au) in case of one short stack. The arrangement is shown in Fig. 9.

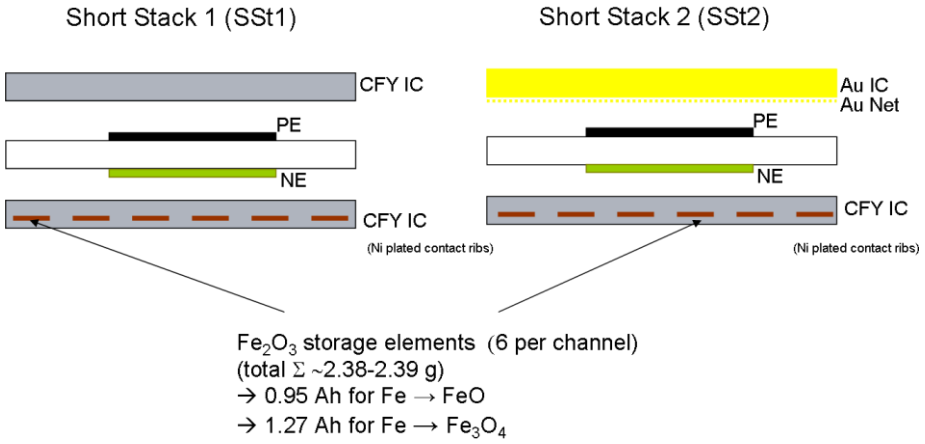


Figure 9. Short stack configuration for the 10,000 cycle test.

The test ended successfully mid of Oct. 2012 at more than 10,000 cycles. Each cycle had a period of about 30 min.. The short stack with CFY as top plate has to be readjusted after some 1,500 cycle in its voltage limits but showed then nearly the same behavior like the stack with the Au-top plate. Even after 10,000 cycles, nearly 50% of the storage material could be discharged (see Fig. 10 which is a snapshot at the time of publishing).

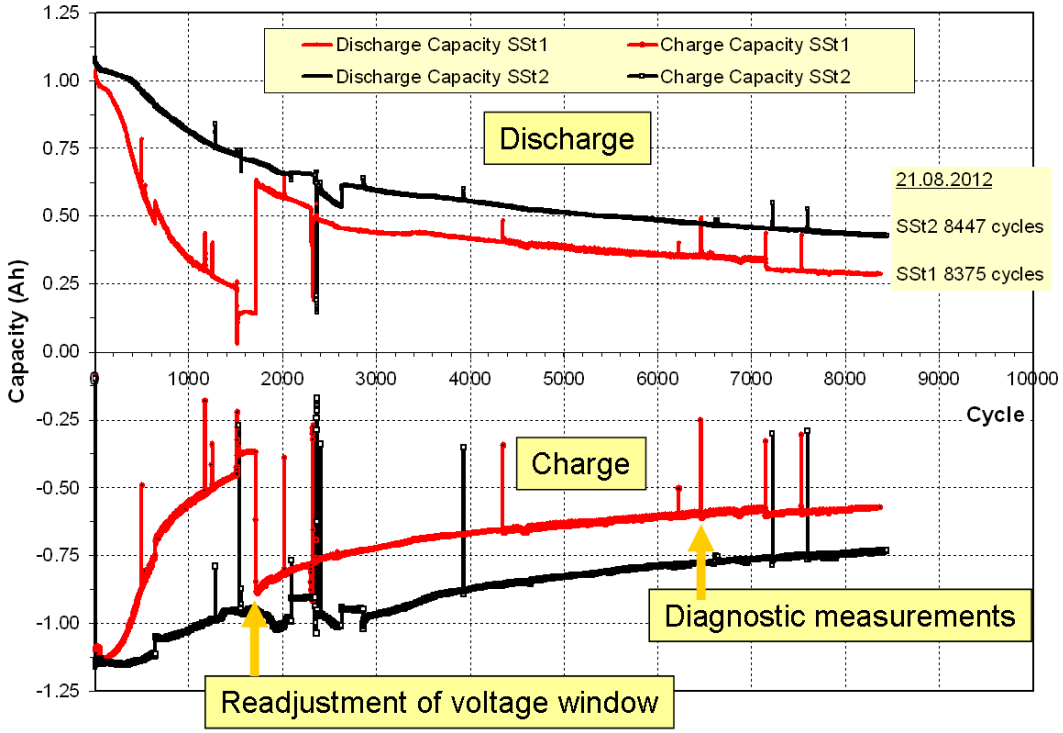


Figure 10. Storage capacity degradation. - Snapshot at the time of publishing.

For an ESC MEA/CFY IC plate configuration it was the first time demonstration of 10,000 Redox-cycles.

Resume: The initial tests of the long term behavior of a metal-air battery operated at 700-800°C have been very successful. All battery components can be optimized with respect to this application, but the results gained up to now lead to an optimistic view for the future of this new type of battery.

6. Systems engineering

The main features of the battery system are

- Integration of the battery into the customer processes,
- to provide for the functionalities requested by the customer within the guaranteed battery performance boundaries,
- to ensure safe and reliable operation of all battery-internal components and the system as a whole within the specified availability limits,
- to monitor the internal battery status in order to allow for a reliable forecast of performance parameters within the operator's planning horizon.

The major system task is to engineer a solution for a thermodynamic challenge inherent to this new type of battery technology: the mismatch of the heat balance between the charging process (which is endothermic) and the corresponding heat excess upon discharge. Ways to optimize heat management became one of the key project requirements, i.e. battery system configurations (in the MW range) had to be developed that include a thermal storage device.

The thermodynamic heat balance is superimposed by irreversible heat produced mainly within the electrochemical cells. Such irreversible processes lead to an excess of heat during a complete charging / discharge cycle which may be used to compensate heat losses dissipated from the total system to the environment. During the whole development program several test article size were considered. Each test article will incorporate some of the basic design features that will be used in the construction of future higher power demonstration units.

System engineering was carried out for a 5 kW, a 25 kW and a 100 kW test article which lead finally to the system integration and design of a 1 MW Battery-Plant. Several conceptual designs were evaluated for a 1MW energy storage system. It was identified that the thermal management of the system was a key aspect of the design since it was critical to performance and cost. Specifically, thermal management was required to maintain a nominal system operating temperature ($T_{nominal}$ = 700°C) and a cell temperature gradient ΔT_{cell} = 100°C. Several thermal management concepts were evaluated including recuperator, regenerator and recirculation system concepts.

A lumped parameter thermal network model of the battery stack was developed to investigate the thermal behavior in the steady state and under transients for the various options of thermal management considered.

The regenerator concept required separate, e.g. underground containment chambers that housed the bed of alumina pellets used as thermal storage (see Fig. 11).

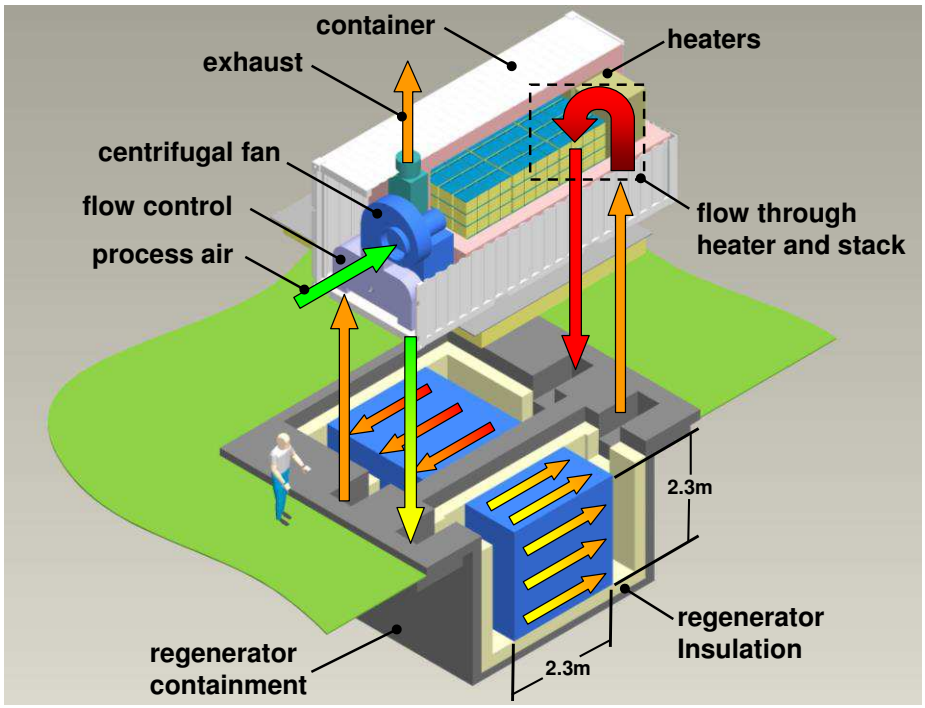


Figure 11. 1MW System with a Regenerator Thermal Management Concept.

A fully integrated system concept was developed for the recirculation approach. The system included a small recuperator, a high temperature centrifugal fan, and an electric air heater. A flow diagram schematic of this system depicting charge and discharge modes is shown in Fig. 12.

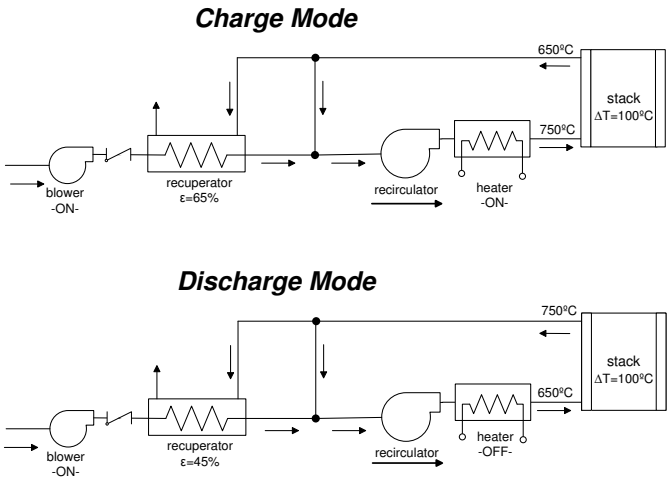


Figure 12. Schematic of Recirculation System Concept.

The design concept for the recirculation system is shown in Fig. 13. Ultimately, the recirculation thermal management system concept was selected based on performance, reliability and cost.

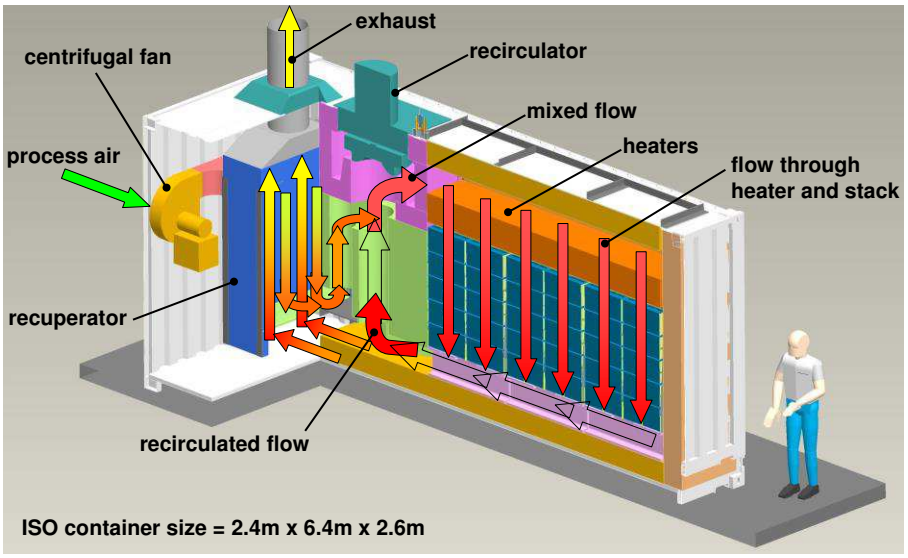


Figure 13. 1 MW System w/ Recirculation Thermal Management Concept.

Summary

After 3 years of developing a novel high temperature metal-air battery the functionality of such an energy storage could be demonstrated. All requirements set have been fulfilled. The system efficiency is better than 60%. Cycle performance of 250-300 mW/cm² over 2 hours was achieved, and the 10,000 load cycles could be demonstrated. The new developed storage material consisting of Fe and Ceramic particles showed an impressive long-term stability and has the potential to become a commercially competitive energy storage system.

Acknowledgments

A wonderful team worked on the project "Novel High Temperature Metal-Air Battery" over 3 years at Siemens Energy, Erlangen, Germany: Wolfgang Drenckhahn (CTO), Michael Kühne, Thomas Soller, Uwe Lenk, Wolfgang Menapace (Head of project);
at Siemens Energy Inc., Pittsburgh, USA: Kevin Litzinger, Jim Shull, Arun Iyengar, Charles Osborne, Chun Lu, William Liu, Gong Zhang, et al.;
at Siemens Corporate Technology, Erlangen and Munich, Germany:
Harald Landes, Horst Greiner, Roland Reichenbacher, André Leonide, Walter Preidel, Katrin Benkert, Harald Schmidt, Alexander Tremel, Carsten Schuh, Marco Cologna, Iris Hahn, Thomas Berthold et al.; all together about 40 co-workers. Thanks to all!

References

1. H. Landes, et al., presented at 222nd ECS meeting, ECS Trans (2012 – 2013).